

Putting the Pieces Together: Deep Learning for Knee MRI Multitissue Abnormality Detection and Severity Grading

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In most radiology research that involves artificial intelligence (AI) applied to image interpretation tasks, deep learning models are trained and tested for binary classification of abnormalities, segmentation of anatomic structures, or localization of disease. These models are applied most commonly to single organ systems or anatomic structures. Although interesting as proofs of concept, most of these AI solutions, as they currently exist, provide only individual pieces of the puzzle of diagnostic image interpretation.

When radiologists interpret imaging studies, they analyze the information in a multifaceted manner. First, there are typically many different organ systems and anatomic structures captured in the imaging study, each of which needs to be analyzed. Second, if any abnormality is present, the radiologist makes a diagnosis, though there may be a long tail of differential diagnoses with different levels of uncertainty. Third, for any diagnosis rendered, there may be a spectrum of disease severity, with important clinical implications. Fourth, if there are prior imaging studies, the findings on the study of interest must be interpreted with consideration of change or stability over time. Fifth, the imaging findings must be interpreted in the context of the patient's clinical history and other nonimaging data. These

different facets of image interpretation are also not independent from each other. AI solutions that address more than one of these facets in combination will be important for creating more useful and clinically valuable AI models.

In this issue of *Radiology: Artificial Intelligence*, Astuto et al (1) put together multiple deep learning puzzle pieces for the task of AI-assisted knee MRI interpretation, specifically addressing the first and third facets described above. They developed a hierarchical three-dimensional (3D) convolutional neural network model to classify knee MRI abnormalities for multiple tissues divided into anatomic compartments, including cartilage (six compartments), bone marrow (six compartments), meniscus (four compartments), and the anterior cruciate ligament. The model was designed to segment relevant anatomic regions of interest, detect abnormalities for each of those regions, and provide a severity class for each abnormality, if detected, based on a simplified version of the Whole-Organ MRI Scoring method (2). The model was trained, validated, and tested using data from 1435 knee MRI studies with 3D fast spin-echo CUBE sequences. They found areas under the receiver operating characteristic curves from 0.83 to 0.93 for classification of binary abnormalities in multiple tissues of interest and also found sensitivities ranging from 43% to 97% for detection of different severity categories for each abnormality.

Although its performance has room for improvement, the AI model provides a framework for an end-to-end AI solution that takes raw knee MRI data as input and produces a relatively comprehensive array of clinically important information. AI models in other radiology subspecialties also have been developed for multiorgan or multitissue abnormality classifications, such as for chest radiographs (eg, numerous analyses using the CheXpert and MIMIC-CXR datasets [3,4]), chest CT (5), and head CT (6), although these models do not include severity grading. The integration of multiabnormality detection with severity grading is a feature that we expect to see in more AI models in the future. However, we also expect to see the incorporation of additional puzzle pieces to address the other facets of diagnostic image interpretation. For example, while the traumatic or degenerative changes typically seen on knee MRI studies are usually not diagnostic dilemmas once identified, for other diseases like musculoskeletal tumors, a differential diagnosis may be necessary. AI models can also be helpful for this task, such as in previous work using deep learning and Bayesian networks to assign differential

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See also the article by Astuto et al in this issue. Conflicts of interest are listed at the end of this article.

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diagnoses on brain MRI studies (7,8). AI models can also be used to compare imaging studies directly and track changes between them (9) and incorporate information from the clinical history and other nonimaging data from the electronic health record (10).

After training their multifaceted knee MRI AI model, Astuto et al (1) also assessed the possible impact of this AI system on study interpretation in a separate test set of 50 knee MRI studies. Two trainees and two attending musculoskeletal radiologists each interpreted the MRI studies first without and then with AI assistance after a washout period, where the AI model information was displayed in a graphical interface. Importantly, accounting for the complexity the multitissue and multiclass outputs provided by their AI model, the authors carefully considered how the AI data would be presented to the people interpreting the studies. They created a graphical schematic with color coding to facilitate the communication of this information. Such design considerations will be important for the clinical deployment of AI models. From this analysis, they found that the interrater agreements between the trainees and attending radiologists improved for the classification of many knee MRI abnormalities when using their AI system. While the AI system did improve the interrater agreement between trainees and musculoskeletal radiologists, in routine clinical practice, trainees are not providing the final interpretations on imaging studies. Thus, the more clinically important question is whether or not such an AI system would improve the interrater agreement between attending radiologists in relation to a stronger ground truth (eg, multiexpert consensus). While the results of this study are promising, future research into the impact of such models on radiologist performance is warranted.

Severity grading is an important feature of the knee MRI AI model presented, which could help to accelerate more consistent reporting of knee MRI studies in routine clinical practice. However, while the binary abnormality classification performance was high, the performance for severity category classification was more modest, which could be at least in part due to the weakness of the ground truth labels used in the study. The 1435 knee MRI examinations used for model training, validation, and testing were annotated by one of five radiologists who each labeled nonoverlapping components of the dataset. In the analysis of multiple raters on 50 knee MRI studies with and without use of the AI system, the interrater agreement κ coefficients for two musculoskeletal radiologists without AI assistance ranged from 0.55 to 0.72. Thus, the ground truth assigned by single raters to the 1435 knee MRI dataset likely had substantial variability. The model performance would likely improve with stronger ground truths, ideally from the consensus or average from multiple

imaging experts instead of single raters. These results highlight the importance of high-quality ground truth labels for the development of radiology AI tools, particularly for severity grading where, in comparison to binary classification tasks, there may be higher inter- and intrarater variability of expert raters.

We are excited about the potential of AI models like the one reported in this issue by Astuto et al (1) that put the pieces together to provide a more comprehensive assessment of a diagnostic imaging study. Their multitissue knee MRI deep learning model with severity grading demonstrates the power of such a system, and they show how such a system can improve the consistency of knee MRI study interpretation. The puzzle of diagnostic image interpretation remains far from complete, but this work is a valuable contribution toward the effort to solve it.

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